Benchmarking Forecast Rankings for Deployment-Aware Model Selection

Abstract

Forecasting benchmarks typically rank models by forecast-side metrics. In deployment, however, forecasts are often translated into actions through fixed operational interfaces, and the model preferred by forecast-side evaluation need not be the model that performs best after execution. We study this as a forecasting benchmark question: under a fixed frictional interface, does forecast-side ranking remain reliable for deployed model selection? We focus on Q2, the fixed-interface selection problem, and use Q1 only as mechanism support for why divergence can arise. Across a synthetic zero-friction anchor, a forecasting-native event-micro benchmark, a complementary Traffic-Hourly alert family, and inventory corroboration, we find that forecast-side winners can become recurrently deployed-suboptimal as friction increases. These results do not invalidate forecast-side evaluation for prediction; rather, they suggest that forecasting benchmarks aimed at deployment may need to report deployed-selection robustness alongside forecast-side metrics.

1 Introduction

Forecasting benchmarks typically compare models using forecast-side metrics. This is appropriate when the goal is to measure predictive quality. But many forecasting systems are not used directly: they are deployed through fixed interfaces that translate forecasts into actions under operational frictions such as switching costs, inertia, or execution constraints. In such settings, the model selected by forecast-side evaluation need not be the model that performs best after execution.

This paper studies that mismatch as a forecasting evaluation problem. Our main question is not how to learn a better policy, but whether forecast-side model ranking remains reliable for deployment-aware model selection under one fixed forecast-to-decision interface. We call this Q2. We separate Q2 from a mechanism question, Q1, which asks whether the same proposed action path can produce different realized outcomes under different interfaces. This separation keeps forecast-side metrics in their role as prediction measures while isolating when the object of model selection changes under deployment friction.

Forecasting benchmark work often evaluates forecast quality, calibration, and live predictive performance. Our question is complementary: whether forecast-side ranking remains trustworthy for deployed model selection under a fixed frictional interface. Unlike decision-aware forecasting or control work, we do not propose new training objectives or policies; we study the evaluation object used for model selection after forecasts are mediated by actions and friction.

This paper makes three narrow contributions.

- First, we introduce deployed-selection robustness as a forecasting benchmark question: whether forecast-side model ranking remains reliable after forecasts are executed through a fixed interface.
- Second, we document a Q2 failure mode: the forecast-side winner becomes recurrently deployed-suboptimal as friction increases under the studied fixed interfaces.
- Third, we provide a Q2-first evidence hierarchy spanning synthetic, forecasting-native event and traffic evidence, and operational corroboration; Q1 is retained only as mechanism support.

2 Two Evaluation Questions

We distinguish three evaluation objects: forecast-side quality, target-side decision quality, and realized-action quality. Forecast-side quality measures predictive performance, target-side decision quality scores the proposed action path before frictional execution, and realized-action quality scores the executed path after the fixed interface is applied. Q1 holds the proposed action path fixed and varies only the interface, so it is an exact-control mechanism check. Q2 holds the interface fixed and varies the forecaster, so it is the benchmark-selection question. Because Q2 asks whether the forecast-side winner remains the deployed winner, Q2 is the paper's main empirical claim.

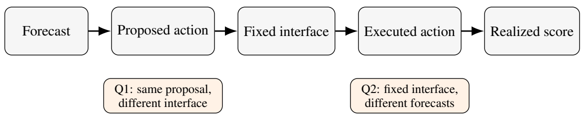

Figure 1: Forecast-to-decision evaluation setup. Forecasts are converted into proposed actions and then executed through a fixed frictional interface. Q2, the paper's main question, asks whether forecast-side model ranking matches deployed model ranking under one fixed interface. Q1 is used only as mechanism support.

3 Evidence Design

Our evidence hierarchy is Q2-first. Synthetic provides an exact zero-friction sanity anchor. Event-micro provides the main forecasting-native Q2 evidence: forecasters emit event probabilities, a fixed threshold maps probabilities to actions, and switching friction penalizes action changes. Traffic-Hourly Top-k provides a second forecasting-native task family under the same fixed-interface selection question. Inventory provides operational corroboration, while load-following and broader inventory sweeps are retained as appendix-only transparency checks. Across domains, the central quantity is whether the forecast-side winner remains the deployed winner under the fixed interface.

In inventory Q2, the main text retains only the 0.50 and

1.00 rows from the fixed five-family comparison; the mixed 0.00/0.25 regime and broader family sweep remain appendix-only transparency checks.

4 Results

4.1 Q2: Fixed interface, benchmark failure and selection consequence

Q2 asks the paper's headline benchmark question: under one fixed forecast-to-decision interface, can forecast-side ranking still be trusted for deployed model selection? Synthetic provides the exact zero-friction sanity anchor; Figure 2 focuses on the fixed-interface winner-inversion evidence in the forecasting-native and corroborative domains.

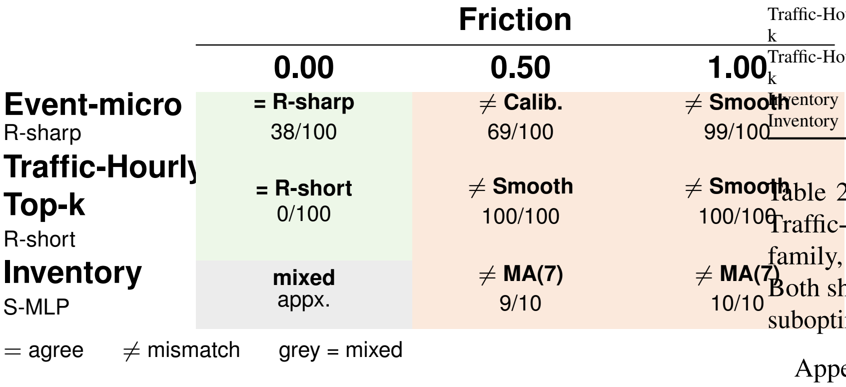

Figure 2: Forecast-side winners need not remain deployed winners. Each cell reports the forecast-side winner and deployed winner under one fixed interface, together with the number of seeds in which the forecast-selected model is deployed-suboptimal. Event-micro and Traffic-Hourly provide forecasting-native evidence; inventory provides operational corroboration at moderate-to-high friction. The inventory zero-friction cell is marked as mixed and is not used as headline corroboration.

In event-micro, the forecast-side winner remains Reactive sharp across frictions, but the deployed winner shifts to Calibrated baseline at friction 0.50 and Lagged smoother at friction 1.00. The benchmark consequence is recurrent: the forecast-selected model is deployed-suboptimal in 69/100 and 99/100 seeds at these two friction levels. The same qualitative inversion persists under an alternate threshold, under a hysteresis-threshold interface, and when forecast-side ranking is recomputed with log loss.

Friction	Forecast winner	Deployed winner	Agree.	Subopt. seeds
0.00	Reactive sharp	Reactive sharp	0.62	38/100
0.50	Reactive sharp	Calibrated baseline	0.31	**69/100**
1.00	Reactive sharp	Lagged smoother	0.01	**99/100**

Table 1: Event-micro Q2 evidence. Under a fixed thresholding interface, the forecast-side winner remains Reactive sharp, but the deployed winner shifts as switching friction increases. The forecast-selected model becomes recurrently deployed-suboptimal at moderate and high friction.

A complementary Traffic-Hourly fixed-budget alert family shows the same benchmark consequence under a different forecasting-native task structure. In the selected Top-k setting ($k = 249$), Reactive short is both the forecast-side winner and the deployed winner at zero friction, with exact agreement 1.0. Once switching costs are introduced, however, the forecast-side winner remains Reactive short while the deployed winner shifts to Lagged smoother, and deployed-suboptimality reaches $100/100$ seeds at both friction 0.5 and friction 1.0. A relative-rank Traffic variant shows the same qualitative pattern in the appendix.

Inventory provides operational corroboration rather than the main forecasting-native evidence. At friction 0.50 and 1.00, the forecast-selected Small MLP is deployed-suboptimal in 9/10 and 10/10 seeds, respectively, under the fixed replenishment interface.

Domain	Friction	Forecast winner	Deployed winner	Subopt. seeds
Traffic-Hourly Top-k	0.00	Reactive short	Reactive short	0/100
Traffic-Hourly Top-k	0.50	Reactive short	Lagged smoother	100/100
Traffic-Hourly Top-k	1.00	Reactive short	Lagged smoother	100/100
Inventory	0.50	Small MLP	Moving average (7)	9/10
Inventory	1.00	Small MLP	Moving average (7)	10/10

Table 2: Forecasting-native and operational corroboration. Traffic-Hourly provides a second forecasting-native task family, while inventory provides operational corroboration. Both show that forecast-side winners can become deployed-suboptimal under fixed interfaces at moderate-to-high friction.

Appendix-only robustness checks preserve the same qualitative reading and do not overturn the narrower main-text claim.

Q1 mechanism support. Q1 explains why divergence can arise; Q2 is the actual benchmark-selection claim. Holding the proposed action path fixed, synthetic and inventory show that frictional interfaces can separate proposed and realized actions. Synthetic yields an increasing proposal-versus-realization gap away from zero friction, while inventory shows a threshold pattern in which sufficiently high friction favors tempered execution. Q2 then shows when that mechanism produces deployed misselection under one fixed interface.

5 Discussion, Related Work, and Limitations

The paper's practical claim is narrow: forecast-side metrics may remain valid for prediction, yet under a fixed deployed interface they may be insufficient for deployed model selection. Event-micro provides the main forecasting-native Q2 evidence, Traffic-Hourly provides a complementary forecasting-native family under the same fixed-interface question, inventory provides operational corroboration, and Q1 explains why the evaluation object can change under friction.

Forecasting benchmarks and leaderboards typically emphasize forecast-side accuracy, calibration, or live question resolution rather than fixed-interface deployed model selection [1, 2, 3]. Recent efforts such as ForecastBench, Prophet Arena, and GIFT-Eval broaden forecasting evaluation through dynamic, live, or zero-shot benchmark design [1, 2, 3]. Our question is complementary: under one fixed operational interface, does the forecast-side winner remain the deployed-best model? Our contribution is evaluative rather than algorithmic: we do not learn a policy or modify the training objective. We contrast with decision-focused learning and control-oriented formulations only to clarify scope: we do not optimize a policy or modify training objectives. Unlike task-based and smart predict-then-optimize methods that change training to improve downstream decisions [4, 5], we keep training fixed and study the evaluation object used for model selection.

Three limitations bound the claim. Forecasting-native Q2 evidence now spans more than one task family, though it is still not a broad benchmark suite. Inventory remains corroborative because its zero/low-friction regime is mixed, while load-following remains appendix-only secondary corroboration for the same reason. In the expanded-family inventory sweep, moderate/high-friction mismatch strengthens but zero- and low-friction rows remain mixed, so we do not use that broader comparison as main-text confirmation.

The resulting implication is evaluative rather than predictive: forecasting benchmarks aimed at deployment may benefit from reporting deployed-selection robustness alongside forecast-side metrics.

References

[1] E. Karger, H. Bastani, C. Yueh-Han, Z. Jacobs, D. Halawi, F. Zhang, and P. E. Tetlock. ForecastBench: A Dynamic Benchmark of AI Forecasting Capabilities. In *ICLR*, 2025. arXiv:2409.19839.

[2] Q. Yang, S. Mahns, S. Li, A. Gu, J. Wu, and H. Xu. LLM-as-a-Prophet: Understanding Predictive Intelligence with Prophet Arena. arXiv:2510.17638, 2025.

[3] T. Aksu, G. Woo, J. Liu, X. Liu, C. Liu, S. Savarese, C. Xiong, and D. Sahoo. GIFT-Eval: A Benchmark for General Time Series Forecasting Model Evaluation. NeurIPS Workshop on Time Series in the Age of Large Models, 2024. Extended version: arXiv:2410.10393.

[4] P. L. Donti, B. Amos, and J. Zico Kolter. Task-based End-to-end Model Learning in Stochastic Optimization. In *NeurIPS*, 2017. arXiv:1703.04529.

[5] A. N. Elmachtoub and P. Grigas. Smart "Predict, then Optimize". *Management Science*, 68(1):9–26, 2022.

A Evidence Hierarchy

Domain	Question	Fixed	Varying	Zero-fric.	Positive-fric.	Role
Synthetic	Q2	interface	forecaster	aligned at zero	ranking flips	zero-fric. anchor
Event micro	Q2	interface	forecaster	mixed at zero	selection failure recurs	main fcst.-native Q2 evid.
Traffic Top-k Alert	Q2	interface	forecaster	aligned at zero	switching-cost divergence recurs	comp. second fcst.-native family
Traffic Relative-Rank	Q2	interface	forecaster	aligned at zero	qualitative separation recurs	appx.-only fcst.-native support
Inventory	Q2	interface	forecaster	mixed at zero	stronger recurrence	main oper. corrob.
Load-following	Q2	interface	forecaster	mixed at zero	directional recurrence	appx.-only sec. corrob.
Synthetic	Q1	proposal	interface	exact agreement	mismatch emerges	mech. support
Inventory	Q1	proposal	interface	coincide at zero	threshold story	mech. support

Table A1: Evidence map for the paper's Q2-first hierarchy, covering the synthetic, event-micro, traffic, inventory, and load-following evidence blocks used in the paper and appendix.

B Q1 Mechanism Support

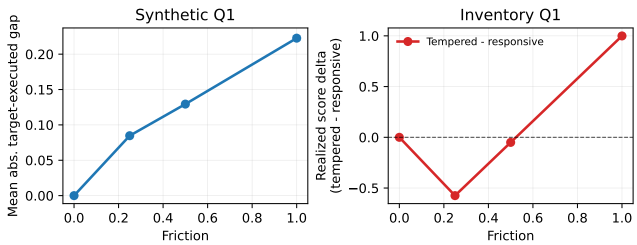

Figure A1: Appendix Q1 mechanism support. Synthetic yields exact zero-friction coincidence and a rising positive-friction proposal-versus-realization gap, while inventory shows a threshold pattern in which sufficiently high friction favors tempered execution. The figure explains why Q2 divergence can arise but is not itself the paper's main evidence.

C Additional Q2 Reliability Checks

This appendix collects the cross-domain seed-level reliability checks and the expanded same-interface inventory comparison. These appendix-only additions do not overturn the narrower main claim from the five-family inventory comparison summarized in the main text.

The same Q2 package also admits simple seed-level reliability checks. Table A3 reports one-sided exact binomial tests for whether the forecast-selected model is deployed-suboptimal in a majority of seeds in the main recurrence rows. Table A4 reports 95% bootstrap intervals for the deployed-gap rows. Figure A2 adds uncertainty bands for the synthetic and event-micro Q2 curves, while Figure A3 shows seed-level disagreement strips for the main Q2 domains.

Domain	Friction	Deployed-suboptimal seeds / total	Share	95% exact CI	One-sided binomial p
Event micro	0.50	69/100	0.69	[0.605, 1.000]	<0.001
Event micro	1.00	99/100	0.99	[0.953, 1.000]	<0.001
Inventory	0.50	9/10	0.90	[0.606, 1.000]	0.011
Inventory	1.00	10/10	1.00	[0.741, 1.000]	<0.001
Load-following	0.25	7/10	0.70	[0.393, 1.000]	0.172
Load-following	0.50	7/10	0.70	[0.393, 1.000]	0.172
Load-following	1.00	8/10	0.80	[0.493, 1.000]	0.055

Table A3: One-sided exact binomial tests for whether the forecast-selected model is deployed-suboptimal in a majority of seeds. The strongest event-micro and inventory rows show clear majority recurrence, while load-following shows the same directional pattern with smaller seed counts.

Domain	Friction	Mean deployed gap	Mean gap 95% bootstrap CI	Median deployed gap	Median gap 95% bootstrap CI
Event micro	0.50	0.011	[0.009, 0.014]	0.008	[0.004, 0.011]
Event micro	1.00	0.057	[0.052, 0.063]	0.053	[0.048, 0.059]
Inventory	0.50	0.308	[0.186, 0.443]	0.227	[0.176, 0.484]
Inventory	1.00	1.187	[0.995, 1.390]	1.162	[0.969, 1.389]
Load-following	0.25	0.001	[0.000, 0.001]	0.000	[0.000, 0.001]
Load-following	0.50	0.001	[0.000, 0.002]	0.001	[0.000, 0.002]
Load-following	1.00	0.002	[0.001, 0.005]	0.002	[0.000, 0.003]

Table A4: 95% bootstrap intervals for deployed-gap rows in the main Q2 evidence blocks. Event-micro and inventory provide the strongest positive-gap support, while load-following remains directional secondary corroboration.

Domain	Role	Status	Seeds	Zero flip	Zero rho	Best +fric.	Best +flip	Strongest pair	Flip share	Note
Synthetic	required	clears stricter gate	20	0.000	1.000	0.2500	0.500	Linear AR / Reactive heuristic	1.000	passed stricter gate
Inventory	required	mixed zero row	10	0.120	0.840	1.0000	0.500	Small MLP / Moving average (7)	1.000	does not clear stricter gate

Table A2: Expanded same-interface inventory comparison summary retained for appendix transparency. These statuses do not overturn the narrower main-text claim from the five-family inventory comparison.

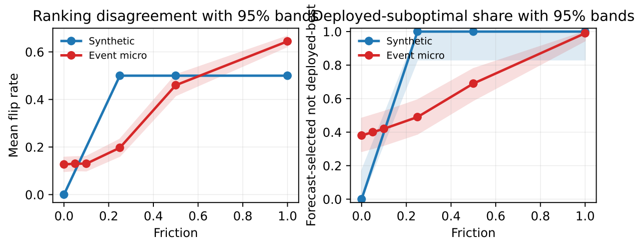

Figure A2: Appendix uncertainty bands for the synthetic and event-micro Q2 curves. Left: mean flip rate with 95% bands from seed-level standard errors. Right: deployed-suboptimal share with exact 95% binomial intervals.

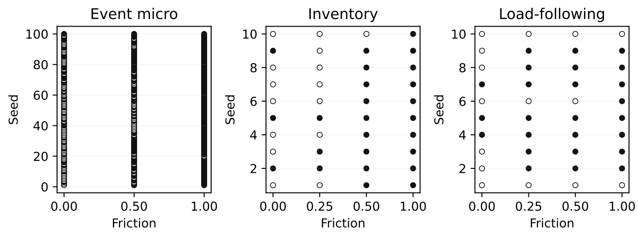

Figure A3: Seed-level disagreement strips for the main Q2 domains. Filled markers indicate seeds where the forecast-selected model is not deployed-best under the fixed interface.

D Event-Micro Robustness Package

Because event-micro provides the paper's main forecasting-native Q2 evidence, this appendix records the first domain-specific robustness package for that benchmark. We construct a minimal forecasting-native binary-event setting evaluated over 100 seeds, in which each forecaster emits event probabilities, a fixed thresholding rule maps probabilities to actions, and switching friction penalizes action changes. The main text uses Brier as the canonical forecast-side criterion. This appendix records the full six-row Brier table plus an alternate-threshold check, a log-loss reranking check, and a second-interface hysteresis check.

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap
0.00	Reactive sharp	Reactive sharp	0.62	0.003
0.05	Reactive sharp	Reactive sharp	0.60	0.002
0.10	Reactive sharp	Reactive sharp	0.58	0.002
0.25	Reactive sharp	Calibrated baseline	0.51	0.003
0.50	Reactive sharp	Calibrated baseline	0.31	0.011
1.00	Reactive sharp	Lagged smoother	0.01	0.057

Table A5: Forecast-side vs. deployed-side model selection in the full six-row event-micro benchmark summary. Under the canonical fixed-threshold interface, deployed-suboptimal counts rise from $38/100$ seeds at zero friction to $99/100$ at friction 1.00.

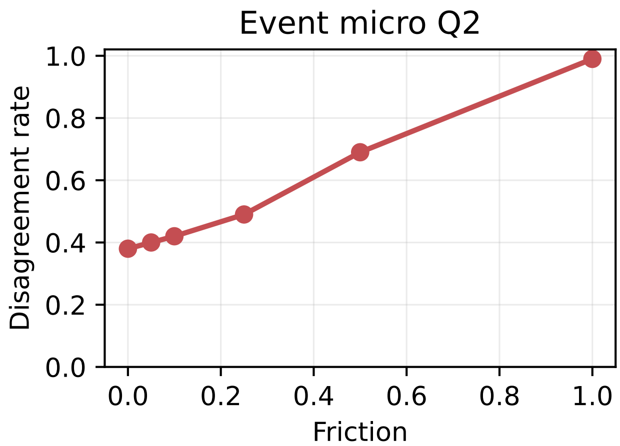

Figure A4: Ranking mismatch in a minimal event-forecasting micro-benchmark. Disagreement between forecast-side and deployed-side model selection is lowest at zero and very low friction and increases as switching frictions rise.

Threshold	Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
tau=0.55	0.00	Reactive sharp	Reactive sharp	0.62	0.003	0.000	38/100
tau=0.55	0.50	Reactive sharp	Calibrated baseline	0.31	0.011	0.008	69/100
tau=0.55	1.00	Reactive sharp	Lagged smoother	0.01	0.057	0.053	99/100
tau=0.50	0.00	Reactive sharp	Reactive sharp	0.56	0.002	0.000	44/100
tau=0.50	0.50	Reactive sharp	Calibrated baseline	0.23	0.014	0.011	77/100
tau=0.50	1.00	Reactive sharp	Lagged smoother	0.02	0.061	0.059	98/100

Table A6: Alternate-threshold robustness for the event-micro benchmark. The same qualitative winner drift and recurrent deployed-suboptimality pattern persists when the fixed threshold is changed from $\tau = 0.55$ to $\tau = 0.50$.

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
0.00	Reactive sharp	Reactive sharp	0.62	0.002	0.000	38/100
0.50	Reactive sharp	Calibrated baseline	0.26	0.013	0.010	74/100
1.00	Reactive sharp	Lagged smoother	0.01	0.060	0.057	99/100

Table A7: Log-loss robustness for the event-micro benchmark at the canonical threshold. The same qualitative winner drift and recurrent deployed-suboptimality pattern persists when forecast-side ranking is recomputed with log loss instead of Brier.

Interface	Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
Hysteresis threshold	0.00	Reactive sharp	Reactive sharp	0.55	0.003	0.000	45/100
Hysteresis threshold	0.50	Reactive sharp	Calibrated baseline	0.45	0.005	0.001	55/100
Hysteresis threshold	1.00	Reactive sharp	Lagged smoother	0.18	0.022	0.020	82/100

Table A8: Second-interface robustness for the event-micro benchmark. Under a fixed hysteresis-threshold interface with $\delta = 0.05$, the same qualitative winner drift and deployed-suboptimality pattern remains visible, though the magnitudes are smaller than under the canonical fixed threshold.

Across this appendix robustness package, the same qualitative winner-drift and recurrent deployed-suboptimality pattern persists in a minimal forecasting-native event-micro benchmark evaluated over 100 seeds.

D.1 Interface-Aware Event-Micro Stats Addendum

Because the hysteresis-threshold check is retained paper-facing as a second-interface robustness result, this addendum records interface-specific recurrence and gap summaries without changing the main cross-domain recurrence package.

Interface	Friction	Deployed-suboptimal seeds / total	Share	95% exact CI	One-sided binomial p
Fixed threshold	0.50	69/100	0.69	[0.605, 1.000]	<0.001
Fixed threshold	1.00	99/100	0.99	[0.953, 1.000]	<0.001
Hysteresis threshold	0.50	55/100	0.55	[0.463, 1.000]	0.184
Hysteresis threshold	1.00	82/100	0.82	[0.745, 1.000]	<0.001

Table A9: Interface-aware recurrence tests for event-micro deployed-suboptimality. The canonical fixed threshold shows stronger recurrence, while hysteresis preserves the same direction with smaller magnitudes.

Interface	Friction	Mean deployed gap	Mean gap 95% bootstrap CI	Median deployed gap	Median gap 95% bootstrap CI
Fixed threshold	0.50	0.011	[0.009, 0.014]	0.008	[0.004, 0.011]
Fixed threshold	1.00	0.057	[0.052, 0.063]	0.053	[0.048, 0.059]
Hysteresis threshold	0.50	0.005	[0.004, 0.006]	0.001	[0.000, 0.003]
Hysteresis threshold	1.00	0.022	[0.019, 0.026]	0.020	[0.015, 0.025]

Table A10: Interface-aware bootstrap intervals for the event-micro deployed gap. Positive intervals remain visible under both interfaces, with smaller gaps under hysteresis.

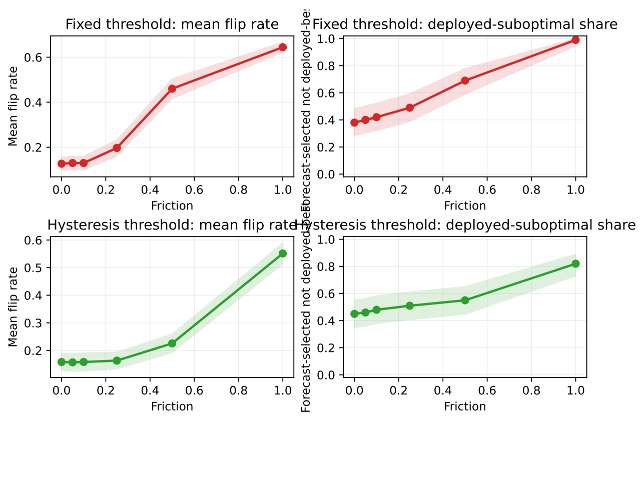

Figure A5: Interface-aware event-micro uncertainty addendum. Rows facet the canonical fixed threshold and the hysteresis threshold; columns show mean flip rate bands and deployed-suboptimal share bands.

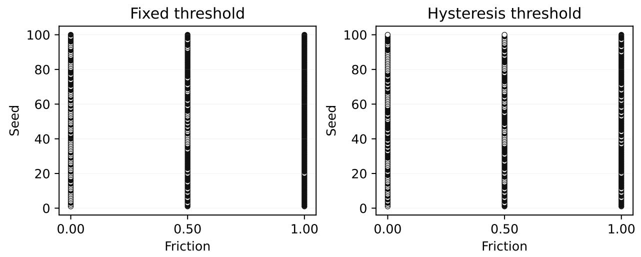

Figure A6: Interface-aware seed-level disagreement strips for the event-micro benchmark. Each panel shows the same seed set under one fixed interface.

E Traffic-Hourly Forecasting-Native Redesign Families

E.1 Fixed-Budget Top-k Alert

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
0.00	Reactive short	Reactive short	1.00	0.000	0.000	0/100
0.50	Reactive short	Lagged smoother	0.00	7.016	6.977	100/100
1.00	Reactive short	Lagged smoother	0.00	24.651	24.473	100/100

Table A11: Forecast-side vs. deployed-side model selection in a complementary fixed-budget Traffic-Hourly alert family. The forecast-side winner is deployment-optimal at zero friction, but switching costs shift the deployed winner to a smoother family.

In the selected Top-k setting, Reactive short is the forecast-side winner and the deployed winner at zero friction. At frictions 0.5 and 1.0, the deployed winner shifts to Lagged smoother even though Reactive short remains Brier-best and continues to win on the forecast side. The qualitative reading is the same as in the main text: under a fixed interface, the more reactive forecast-optimal family need not remain deployment-optimal once switching is penalized.

E.2 Relative-Rank Traffic Variant

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
0.00	Reactive short	Reactive short	1.00	0.000	0.000	0/100
0.50	Reactive short	Calibrated baseline	0.00	10.103	10.122	100/100
1.00	Reactive short	Calibrated baseline	0.00	25.205	25.231	100/100

Table A12: Relative-rank Traffic redesign retained as forecasting-native appendix support. The same qualitative separation between forecast-side and deployed-side selection appears under fixed-budget set action.

The relative-rank target shows the same qualitative separation under a different forecasting-native label construction. Reactive short remains the forecast-side winner, while Calibrated baseline becomes the deployed winner once switching costs are introduced, so we retain this variant as appendix-only forecasting-native support. The alternate $m = 0.15$ variant is also strong in the full report but is omitted from the compact table to keep the appendix narrow.

F Inventory Operational Corroboration

This appendix records the full five-family inventory summary that underlies the compact corroboration rows in the main text. The mixed zero and low-friction rows are retained here for completeness, while the main text uses only the moderate-to-high friction rows to avoid overstating the zero-friction story.

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Deployed-suboptimal seeds / total
0.00	Small MLP	Small MLP	0.70	0.022	3/10
0.25	Small MLP	Linear AR	0.70	0.053	3/10
0.50	Small MLP	Moving average (7)	0.10	0.308	9/10
1.00	Small MLP	Moving average (7)	0.00	1.187	10/10

Table A13: Full inventory Q2 summary retained as appendix corroboration under a fixed replenishment interface.

G Load-Following as a Second Operational Corroboration Domain

We retain a second operational forecasting domain based on aggregated electricity load-following only as appendix-only secondary corroboration. Forecasts are translated into operational actions under a fixed responsive interface, so this section follows the event-micro robustness package as a diagnostic appendix support block rather than as additional main evidence. Using UCI ElectricityLoadDiagrams20112014, we build an aggregate multi-client load-following domain by summing disjoint client groups and evaluating at the selected 60-minute operational resolution. Groups 0 and 1 are used only for restricted calibration, while groups 2–9 serve as held-out evaluation units, so this appendix replaces the earlier household proxy with a stronger cross-sectional operational design.

A grouping-only balance-repair rerun substantially reduces group imbalance and Q1 target clipping while preserving the Q2 drift structure. A final restricted cap/margin search, however, finds no jointly feasible configuration that also satisfies the remaining Q1 requirement, so the domain remains appendix-only secondary corroboration rather than additional Q1 promotion evidence.

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
0.00	Linear AR	Small MLP	0.70	0.000	0.000	3/10
0.25	Linear AR	Small MLP	0.30	0.001	0.000	7/10
0.50	Linear AR	Small MLP	0.30	0.001	0.001	7/10
1.00	Linear AR	Small MLP	0.20	0.002	0.002	8/10

Table A14: Load-following Q2 summary retained as appendix-only secondary corroboration under a fixed responsive interface.

This mixed zero row is not a tie-policy artifact: the zero-row winning sets are singletons in all seeds, yet the selected candidate still shows $3/10$ zero-friction mismatches with mean deployed gap 0.0002 and median gap 0.0. The grouping-only balance repair lowers the zero-row mean gap to about 0.0001 but does not remove the mismatch, so the most plausible reading is near-zero seed instability or grouping granularity rather than a clean zero-friction story.

From friction 0.25 onward, the forecast-side winner remains Linear AR while the deployed winner is Small MLP, agreement stays at 0.30, 0.30, and 0.20, and deployed-suboptimal counts rise to $7/10$, $7/10$, and $8/10$. The domain therefore contributes directionally consistent secondary corroboration only; it does not serve as main evidence or as additional Q1 promotion evidence in the current round.

H Stronger Inventory Q2 Baselines

To reduce baseline-specification ambiguity, we widen the inventory Q2 comparison set with stronger classical and neural families while leaving the environment, split, friction grid, fixed responsive interface, deployed metric, and tie policy unchanged. This appendix section is included as a broader transparency layer rather than as earlier or stronger main evidence than the five-family inventory comparison. All tunable families share the same held-out protocol, validation criterion, and standardized two-stage tuning structure, while the two simple heuristics remain fixed ex ante. Family representatives are selected by validation negative MAE only, and the held-out forecast-side winners in the robustness table are then computed from the same held-out forecast metric used in the main inventory Q2 pipeline.

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Deployed-suboptimal seeds / total
0.00	Small MLP	Gradient-boosted trees	0.60	0.043	4/10
0.25	Small MLP	Gradient-boosted trees	0.60	0.059	4/10
0.50	Small MLP	Gradient-boosted trees	0.50	0.128	5/10
1.00	Small MLP	Moving average (7)	0.10	0.684	9/10

Table A17: Expanded-family inventory Q2 robustness summary for the broader comparison set.

In this broader comparison, agreement is mixed at 0.00 and 0.25, falls further by 0.50 and 1.00, deployed-suboptimal counts rise from $4/10$ to $9/10$, and moderate/high-friction mismatch strengthens. The mixed zero/low-friction rows do not overturn the narrower main claim from the five-family inventory comparison.

I Why Forecast Ordering Need Not Be Preserved

Proposition. A proper forecast-side score need not be order-preserving for deployed utility under a fixed frictional action interface.

Reversal condition. If a forecaster's forecast-side score advantage is smaller than the extra friction-induced utility loss from its induced action path, deployed ordering can reverse.

Proof sketch. Properness concerns truthful prediction scoring. Deployed model selection uses the composed object forecast $\to$ action proposal $\to$ interface $\to$ realized utility. Therefore forecast-side ordering need not be preserved after action mediation.

Family	Input	Stage 1	Stage 2	Seeds	Representative rule
Naive persistence	fixed heuristic	–	–	final 10	fixed ex ante
Moving average (7)	fixed heuristic	–	–	final 10	fixed ex ante
Linear AR	legacy tabular	5	top 5	s1 0,1 / s2 0,1,2 / final 0-9	best mean val. metric, then median, then pre-registered simpler config
Small MLP	legacy tabular	18	top 5	s1 0,1 / s2 0,1,2 / final 0-9	best mean val. metric, then median, then pre-registered simpler config
Small GRU	sequence (lookback 28)	6	top 5	s1 0,1 / s2 0,1,2 / final 0-9	best mean val. metric, then median, then pre-registered simpler config
Regularized linear + lag search	lagged tabular	36	top 5	s1 0,1 / s2 0,1,2 / final 0-9	best mean val. metric, then median, then pre-registered simpler config
Gradient-boosted trees	lagged tabular	216	top 5	s1 0,1 / s2 0,1,2 / final 0-9	best mean val. metric, then median, then pre-registered simpler config
Large MLP	legacy tabular	108	top 5	s1 0,1 / s2 0,1,2 / final 0-9	best mean val. metric, then median, then pre-registered simpler config
GRU variant	sequence	81	top 5	s1 0,1 / s2 0,1,2 / final 0-9	best mean val. metric, then median, then pre-registered simpler config

Table A15: Standardized tuning protocol for the stronger-baseline comparison. All tunable families share the same split, validation criterion, and two-stage selection structure; the two simple heuristics remain fixed ex ante.

Family	Representative	Validation mean	Validation median
Naive persistence	fixed ex ante	-2.911	-2.981
Moving average (7)	fixed ex ante	-3.265	-3.251
Linear AR	alpha=0.0001	-2.449	-2.512
Small MLP	lr=0.001, wd=0.0, batch=32	-2.331	-2.390
Small GRU	lr=0.001, batch=32	-2.628	-2.571
Regularized linear + lag search	lag=7, lasso, alpha=0.1	-2.418	-2.410
Gradient-boosted trees	lag=14, trees=100, depth=5, lr=0.1, leaf=10	-2.425	-2.279
Large MLP	w=256, d=3, drop=0.0, lr=0.001, wd=0.0001	-1.954	-2.012
GRU variant	L=28, h=128, layers=2, drop=0.0, lr=0.001	-2.602	-2.599

Table A16: Validation-selected representatives for the broader inventory Q2 family sweep. Family representatives are chosen by validation negative MAE only.